

Final Project: Network Design and Security Configuration

Lab Report

TechSafe, Ltd. — Network Design and Security Configuration

Date: July 10, 2026

Overview

TechSafe, Ltd. uses a single fixed IP address block from its Internet service provider and asked for a structured network with proper IP addressing, subnetting, and security controls across its Administration, Sales, and IT departments. This report documents the four tasks completed for this project: designing a subnetted network diagram, calculating subnet addresses and masks for each department, capturing and analyzing HTTPS traffic in Wireshark, and configuring Windows Defender Firewall rules to block FTP and HTTP traffic on a workstation.

Environment	Online Windows lab environment
Tools Used	diagrams.net, Wireshark, Windows Defender Firewall with Advanced Security, Command Prompt (ftp), Microsoft Edge / Google Chrome
Tasks Completed	4 of 4 (network diagram, subnet calculations, traffic analysis, firewall configuration)

Task 1: Design a Network with Subnets

Objective: Create a network diagram for TechSafe Ltd. showing one router connected to one switch, with three subnets — Administration, Sales, and IT — connected to the switch.

Using diagrams.net, I built a simple topology with the Router at the top connected to the Switch, and the three department subnets (Administration Subnet, Sales Subnet, and IT Subnet) arranged around the switch and connected to it with solid lines. Each device and subnet is clearly labeled, and no additional network elements such as firewalls or extra switches were added, in line with the task requirements.

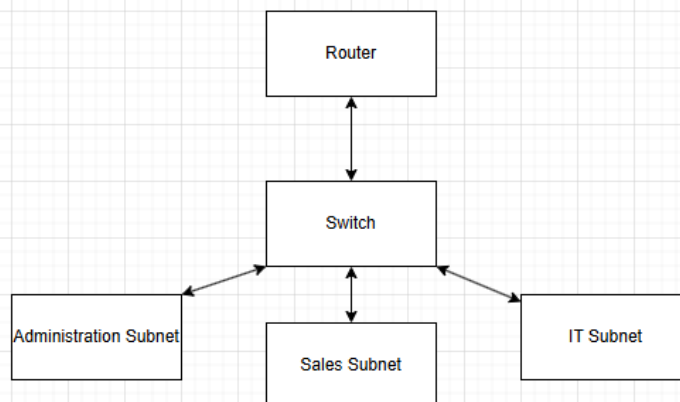


Figure 1. Network diagram showing the Router connected to the Switch, which connects to the Administration, Sales, and IT subnets.

Task 2: Calculate Subnet Addresses and Subnet Masks

Objective: Using the fixed network address 198.51.100.0/24, calculate the subnet address, subnet mask, and IP address range for each of the three departments based on their assigned host ranges.

Each department's range spans 62 usable host addresses, which corresponds to a /26 (255.255.255.192) subnet mask. I calculated the subnet address for each department by identifying the network address that the given host range falls within, and recorded the results in a four-column table with the subnet name, subnet address, subnet mask, and IP address range.

Subnet Name	Subnet Address	Subnet Mask	IP Address Range
Administration	198.51.100.0	255.255.255.192	198.51.100.1 - 198.51.100.62
Sales	198.51.100.64	255.255.255.192	198.51.100.65 - 198.51.100.126
IT	198.51.100.128	255.255.255.192	198.51.100.129 - 198.51.100.190

Figure 2. Table of calculated subnet addresses and subnet masks for the Administration, Sales, and IT departments.

Summary of the calculated values:

Subnet Name	Subnet Address	Subnet Mask	IP Address Range
Administration	198.51.100.0	255.255.255.192	198.51.100.1 - 198.51.100.62
Sales	198.51.100.64	255.255.255.192	198.51.100.65 - 198.51.100.126
IT	198.51.100.128	255.255.255.192	198.51.100.129 - 198.51.100.190

Task 3: Analyze HTTP/HTTPS Website Traffic Using Wireshark

Objective: Capture live network traffic while browsing a selected website, then filter the capture in Wireshark to isolate HTTPS traffic related to that site.

I selected Wikipedia for this task. With Wireshark running, I browsed Wikipedia for over two minutes to generate live traffic, then applied the filter `tcp.port==443` and `tcp contains "Wikipedia"` to isolate HTTPS packets associated with the site. The filtered results show a series of TLSv1.2 Application Data and TCP acknowledgment packets exchanged between my workstation (192.168.1.147) and Wikipedia's servers, confirming an active encrypted session over port 443.

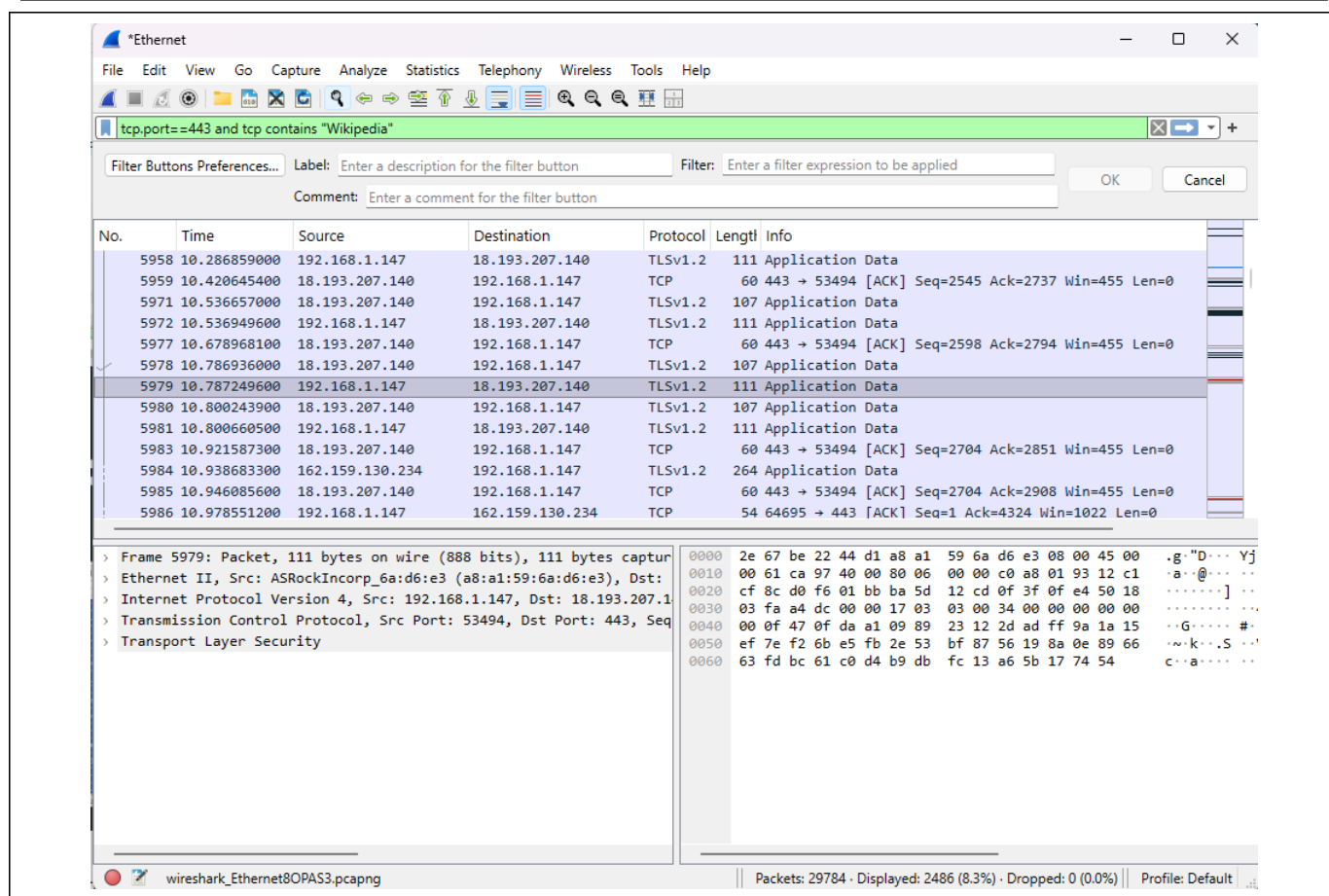


Figure 3. Wireshark capture filtered with `tcp.port==443` and `tcp contains "Wikipedia"`, showing encrypted HTTPS traffic between the workstation and Wikipedia.

Task 4: Configure Firewall Rules Using Microsoft Windows Defender Firewall

Objective: Confirm baseline FTP and HTTP access, then configure outbound rules in Windows Defender Firewall with Advanced Security to block FTP traffic (TCP port 21) and HTTP traffic (TCP port 80), and verify that each rule successfully blocks the corresponding traffic.

Step 1 — Confirm FTP Access Before Applying the Rule

From a Command Prompt, I ran `ftp ftp.dlptest.com` and confirmed a successful connection. The terminal shows the server greeting ("220 Welcome to the DLP Test FTP Server") and a prompt for a username, confirming that outbound FTP traffic was permitted before any firewall rule was applied.

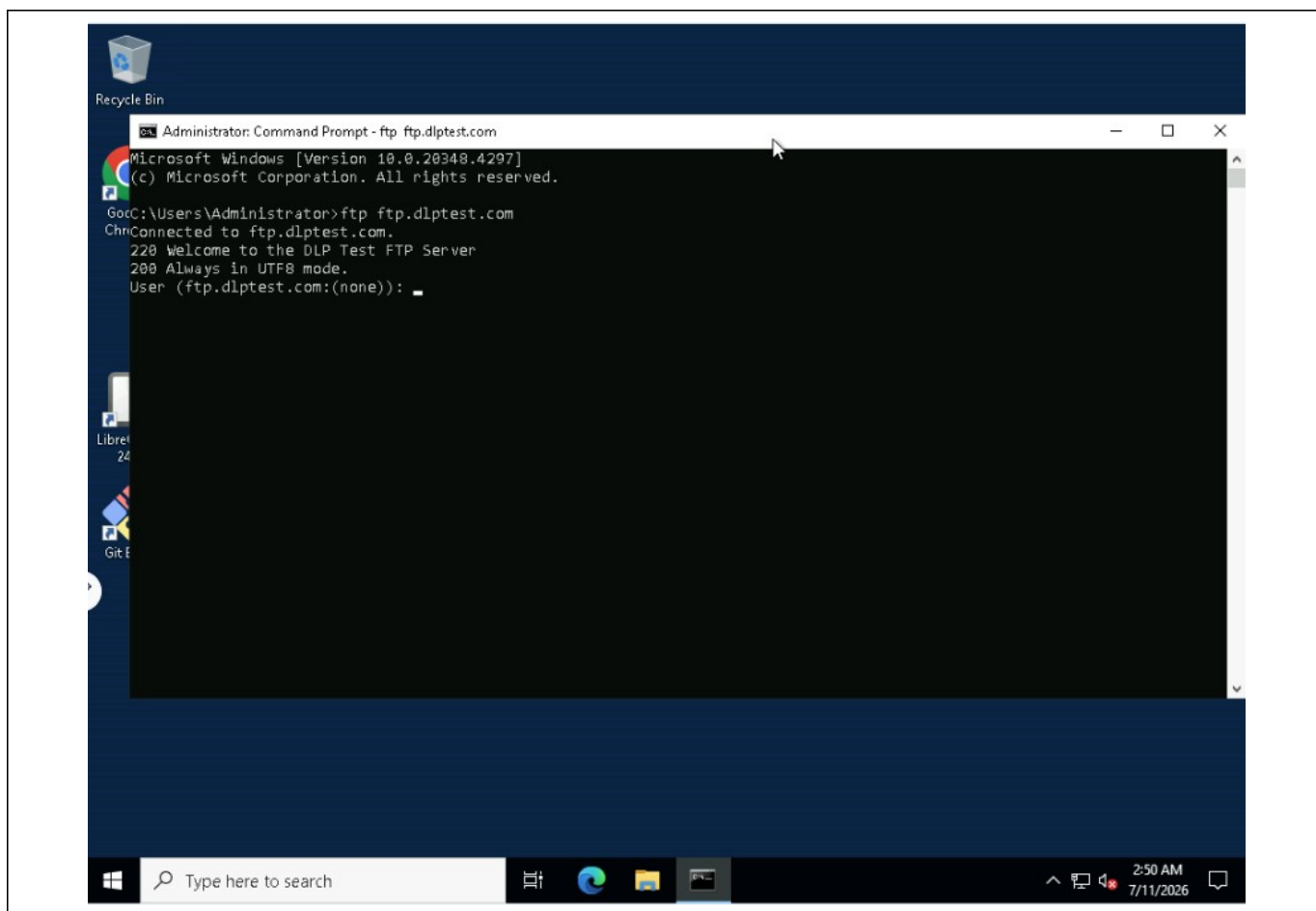


Figure 4. Successful FTP connection to `ftp.dlptest.com` before the firewall rule was applied.

Step 2 — Create an Outbound Rule to Block FTP (TCP Port 21)

In Windows Defender Firewall with Advanced Security, I created a new outbound rule using the Port rule type, selected the TCP protocol, and set the remote port to 21. The rule's action was set to Block, as shown in the Protocols and Ports tab of the rule properties below.

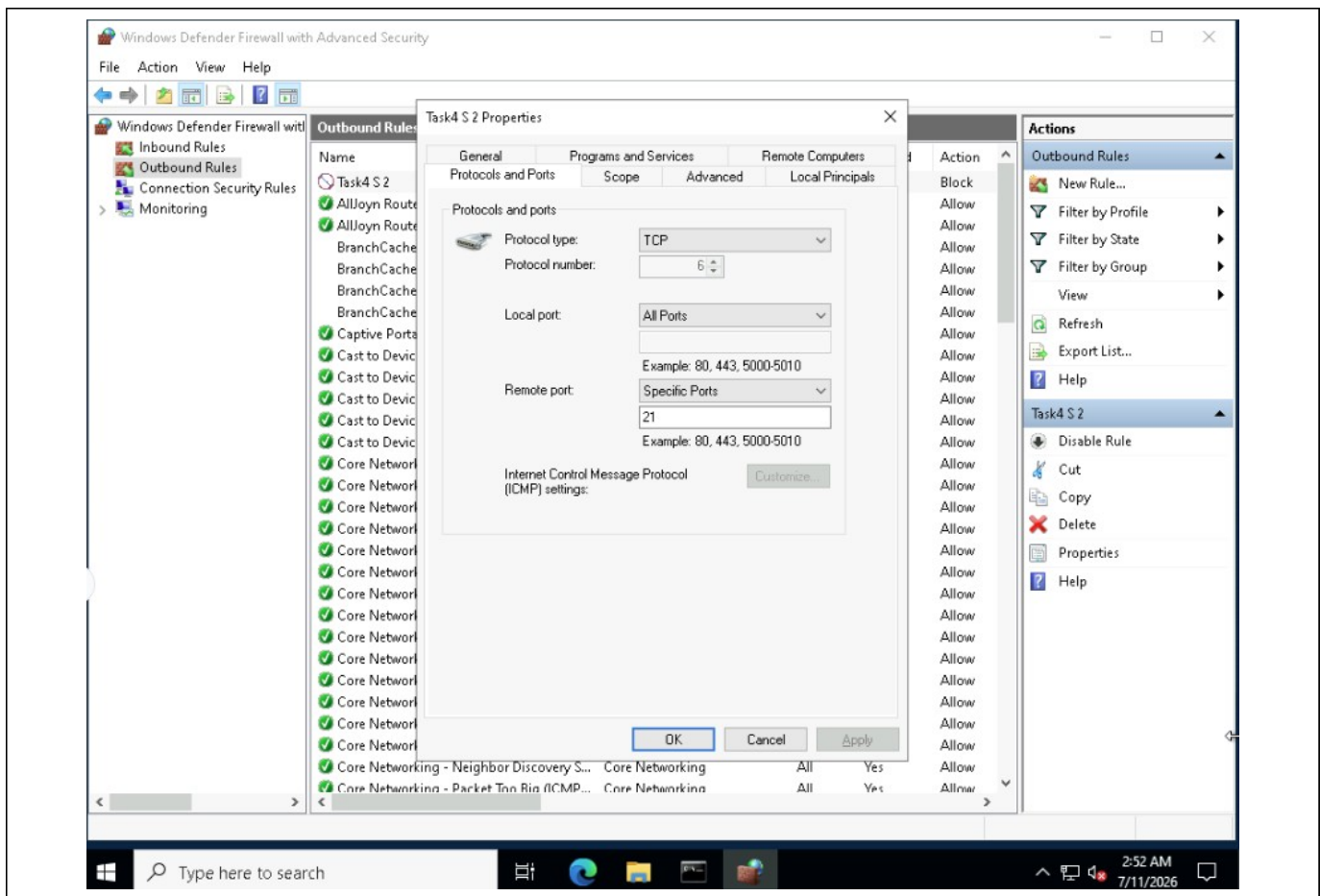


Figure 5. Outbound rule properties for the new FTP-blocking rule, showing TCP protocol with remote port 21 set.

Step 3 — Verify FTP Access Is Blocked

After applying the rule, I ran `ftp ftp.dlptest.com` again from the terminal. Unlike the first attempt, the connection no longer reached the server greeting or username prompt, confirming that the outbound rule successfully blocked FTP traffic on port 21.

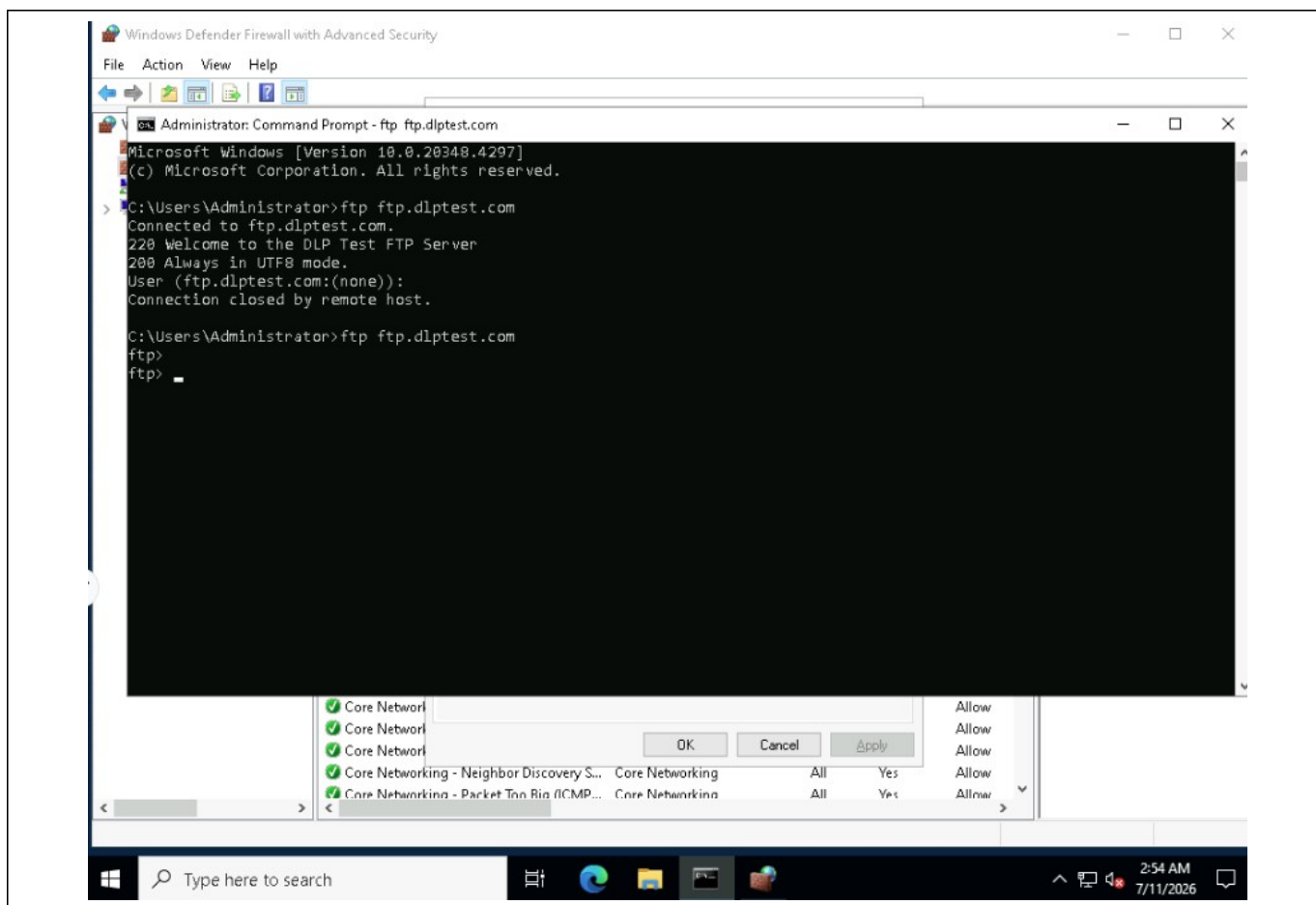


Figure 6. Second FTP connection attempt after the rule was applied, showing the connection no longer completing.

Step 4 — Confirm HTTP Access Before Applying the Rule

Before blocking HTTP traffic, I opened a browser and navigated to <http://httpforever.com/>, a site intentionally served over plain HTTP. The page loaded successfully with a "Not secure" warning in the address bar, confirming that outbound HTTP traffic on port 80 was still permitted at this point.

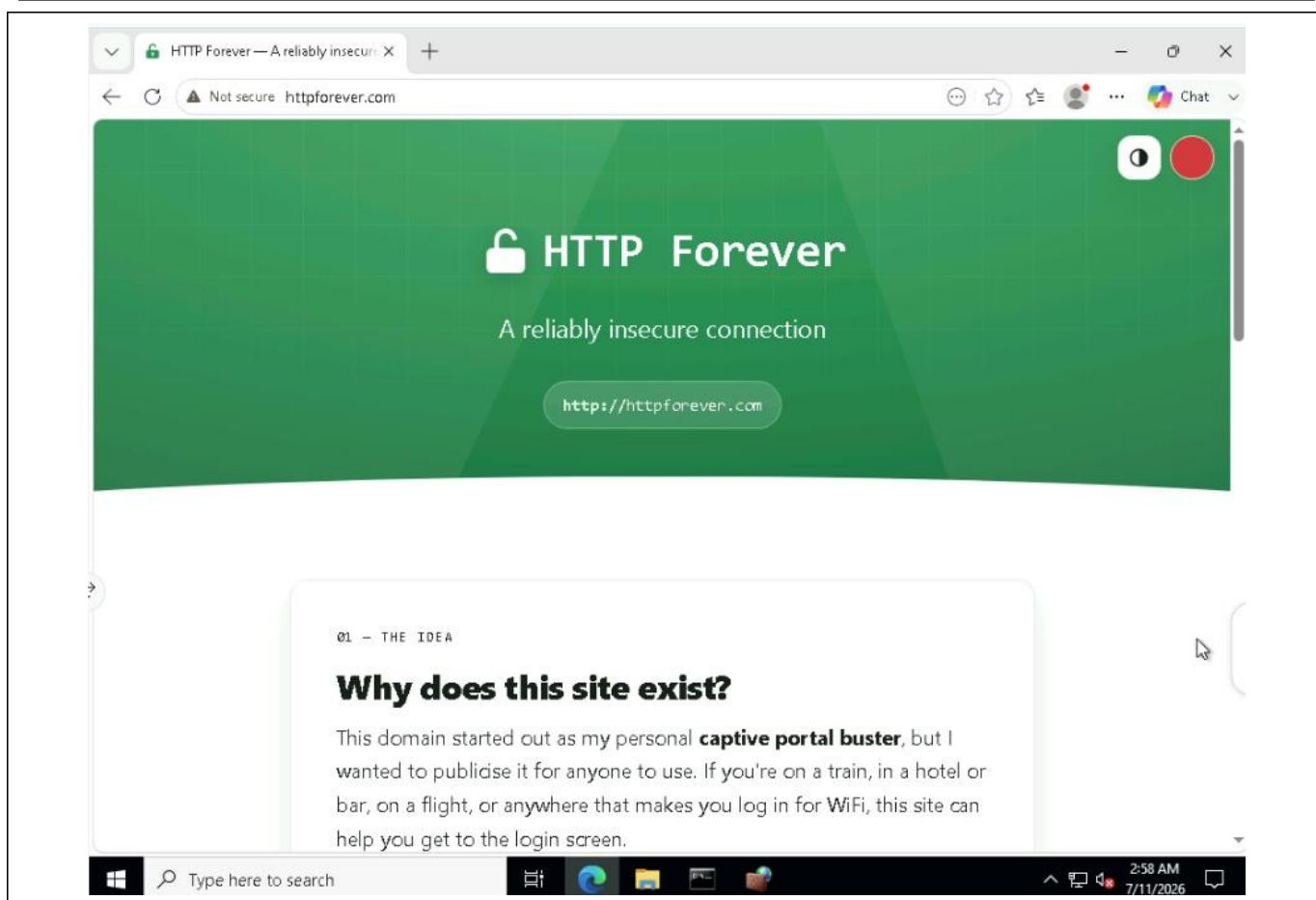


Figure 7. Successful, unblocked access to <http://httpforever.com/> before the HTTP-blocking rule was applied.

Step 5 — Create an Outbound Rule to Block HTTP (TCP Port 80)

I created a second outbound rule in Windows Defender Firewall with Advanced Security, again using the Port rule type with the TCP protocol, this time setting the remote port to 80 and the action to Block.

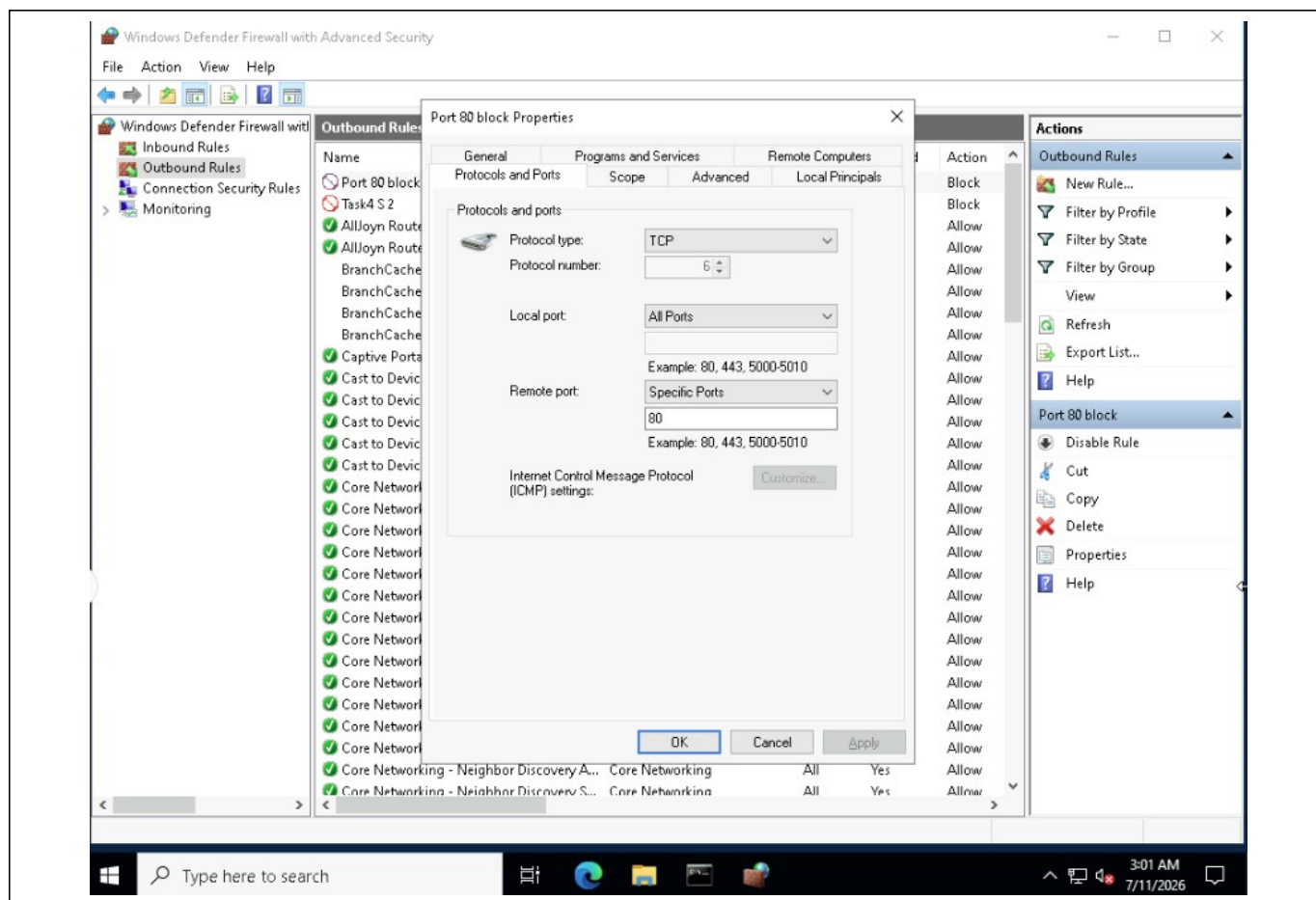


Figure 8. Outbound rule properties for the new HTTP-blocking rule, showing TCP protocol with remote port 80 set.

Step 6 — Verify HTTP Access Is Blocked

After applying the port 80 rule, I attempted to reload <http://httpforever.com/>. The browser returned an "ERR_NETWORK_ACCESS_DENIED" error with the message "Your Internet access is blocked," confirming that the outbound rule successfully blocked HTTP traffic on port 80.

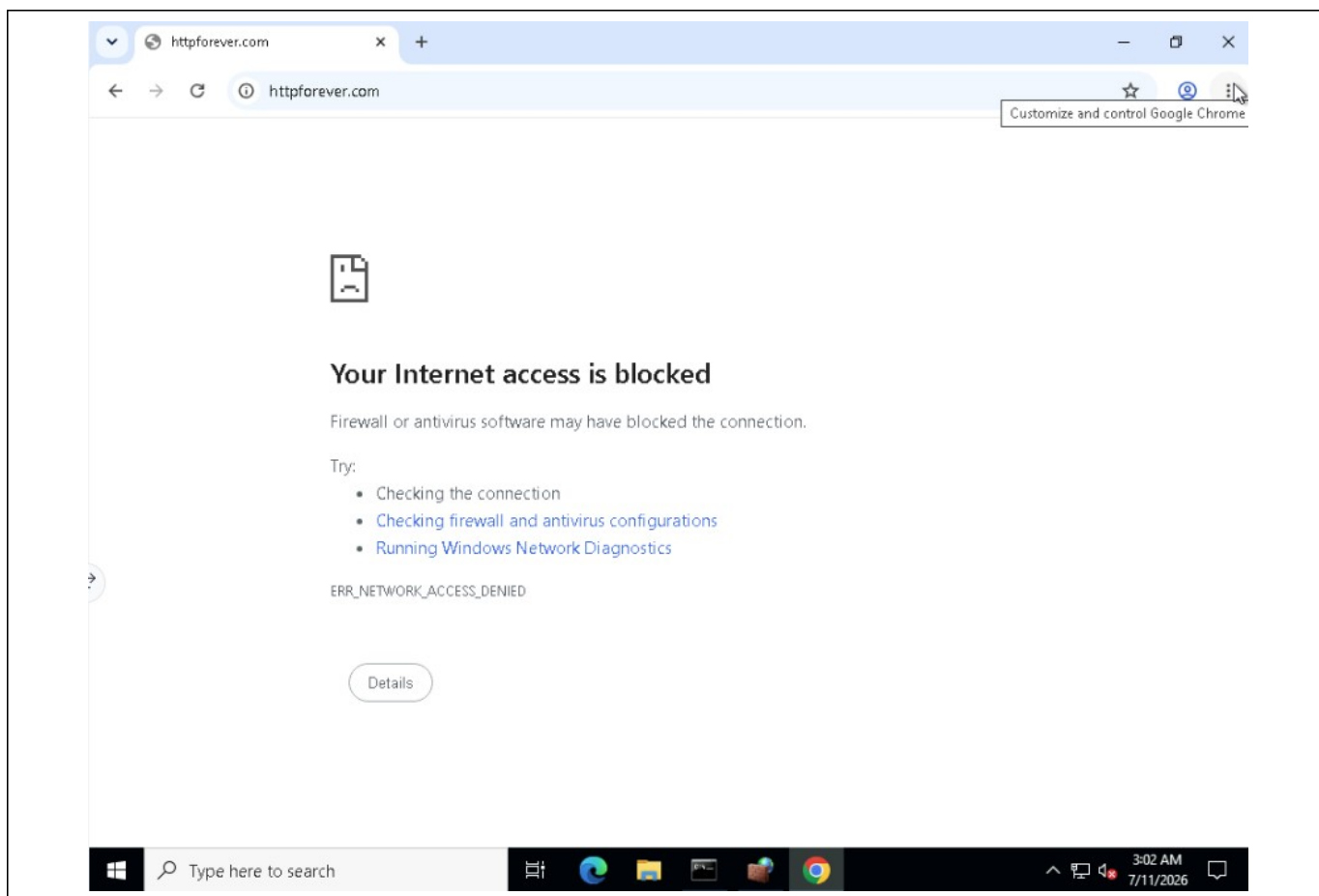


Figure 9. Browser error confirming HTTP access to <http://httpforever.com/> was blocked after the port 80 rule was applied.

Conclusion

All four tasks for TechSafe, Ltd.'s network design and security configuration project were completed successfully. I designed a network diagram with one router, one switch, and three department subnets; calculated the subnet addresses, subnet masks, and IP address ranges for the Administration, Sales, and IT departments based on their assigned host ranges; captured and filtered live HTTPS traffic to Wikipedia in Wireshark to confirm encrypted communication over port 443; and configured, then verified, outbound Windows Defender Firewall rules blocking FTP traffic on port 21 and HTTP traffic on port 80. Together, these tasks demonstrate core networking and security skills, including subnet design and IP addressing, network traffic analysis, and host-based firewall configuration.